



COST FINDING REPORT

FoodShare BD — Food Waste Donation Management System

Submitted by

Name	ID	Contribution
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1. Introduction

This report updates the FoodShare BD cost findings to include a revenue-generating business model, so the platform can fund its own growth rather than relying solely on grants or donations. It follows the Problem Definition, Cost of Development, and Feasibility framework from the CSE346 lecture, and adds an Investment Analysis using the Present Value method ($P = F/(1+I)^n$) to test whether the added investment pays off.

2. Problem Statement

The original FoodShare BD concept matches surplus food from donors to NGOs, shelters, and individuals at no charge to either side. While this serves the social mission, it leaves the platform financially dependent on outside funding, with no way to cover its own hosting, support, and maintenance costs as it scales. A monetization layer is needed that funds operations without charging the recipients the platform exists to help.

3. Project Objectives

- Keep the core donation-matching flow free for individual donors and all recipients (NGOs, shelters, individuals).
- Introduce paid, opt-in value for business donors (restaurants, hotels, event organizers, corporates) who can afford to pay for tangible benefits.
- Reach positive cash flow within a reasonable payback period, verified through a discounted cash flow analysis.
- Keep the added technical scope (payments, subscriptions, reporting) proportionate to a small-team build.

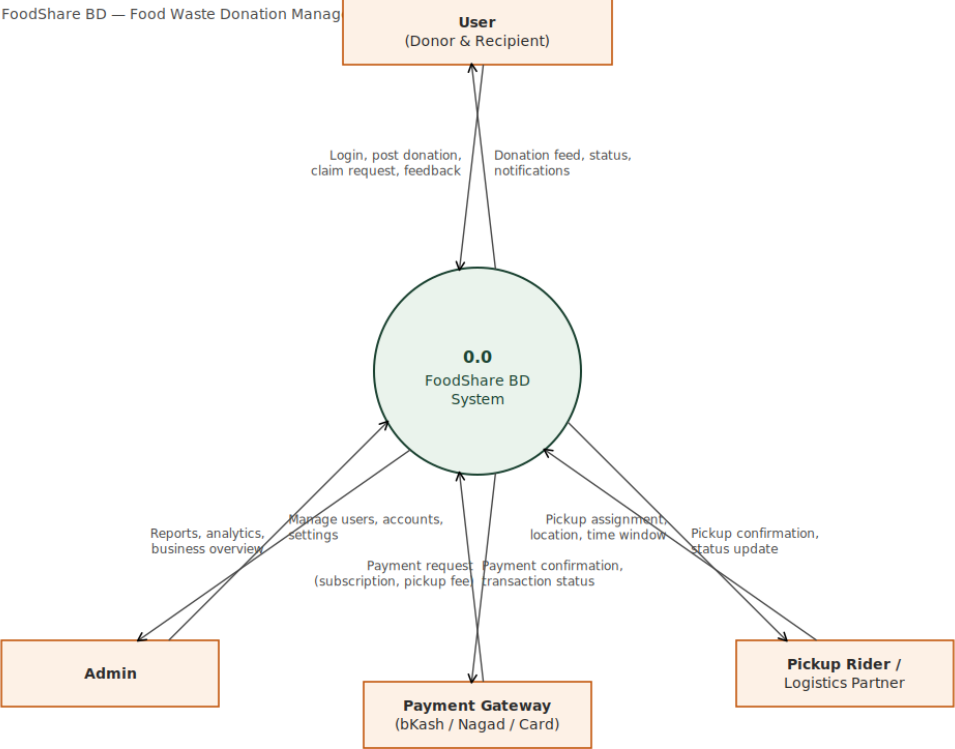
Required Data

The system requires the following data:

- User Profile Information
- Login Credentials
- Waste Food Details
- Uploaded Images
- AI Verification Results
- Pickup Information
- Reward Point Records
- NGO Information
- Activity Logs and Report

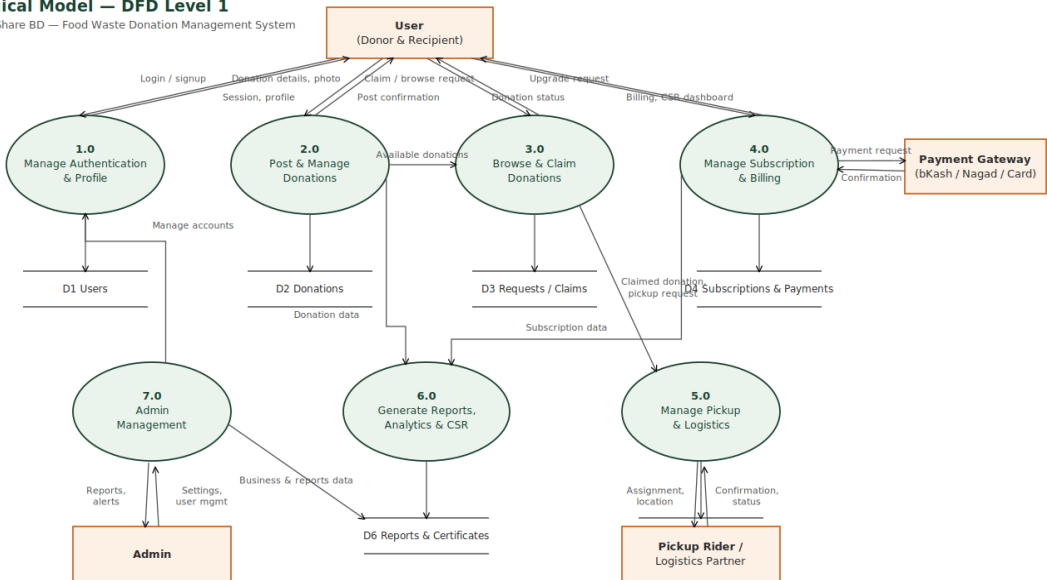
Logical Model — DFD Level 0 (Context Diagram)

FoodShare BD — Food Waste Donation Management System



Logical Model — DFD Level 1

FoodShare BD — Food Waste Donation Management System



4. Business Model

Monetization is aimed at business donors, who gain measurable value from participating, while the platform stays free for individual donors and every recipient — preserving the social mission.

- **Business Pro Subscription — 10,000 Tk./year:** for restaurants, hotels, and event organizers. Unlocks a CSR impact dashboard, downloadable donation certificates for tax/ESG reporting, and priority NGO matching.
- **Corporate CSR Partnership — 60,000 Tk./year per account:** for chains and larger corporates. Adds multi-branch management, a "Verified Partner" badge, dedicated support, and custom sustainability reports.
- **Pickup-as-a-Service fee — 75 Tk./request:** an optional paid, rider-coordinated pickup for donors or NGOs who'd rather not arrange transport themselves. Self-pickup stays free.

Individual donors and all recipients continue to use the core matching features at no cost — only the value-added business services are paid.

5. Project Scope

5.1 Core Functions

- Onboarding and role-based sign-up/login (Donor / Recipient)
- Home/dashboard feed of nearby available donations
- Post-a-donation flow (item details, quantity, photo, pickup window, location)
- Donation detail view, request/claim action, status tracking, and history
- User profile and settings

5.2 Business Model Features

- Business/Restaurant account tier with subscription management (upgrade, billing, renewal)
- Payment integration (bKash / Nagad / card via an aggregator) for subscriptions and logistics fees
- CSR impact dashboard and downloadable donation certificates (PDF)
- Optional paid "Scheduled Pickup" toggle inside the existing post-a-donation flow
- Corporate Partner account type with "Verified Partner" badge shown on listings
- Admin panel additions: manage business/corporate accounts, subscriptions, and logistics assignments

5.3 Out of Scope

- Payments for the core donation itself — donations stay free between donor and recipient; only business services are paid
- Multi-language support beyond English/Bangla text labels
- Web (desktop) client — mobile only for the initial release

6. Cost of Development(Preliminary Solution)

6.1 Alternative 1: Cross-Platform Development (Flutter) — Base App

One-Time Cost Item	Amount (Tk.)
UI/UX Design (Figma — completed)	15,000
Frontend App Development (Flutter, Android + iOS)	80,000
Backend & API Development (Node.js/Express)	70,000
Database Setup (Firebase / PostgreSQL)	15,000
Testing & QA	15,000
Deployment (Play Store + App Store listing)	8,000
Total One-Time Cost	203,000

6.2 Alternative 2: Native Development (Android + iOS separately)

One-Time Cost Item	Amount (Tk.)
UI/UX Design (Figma — completed)	15,000
Android App Development (Kotlin)	90,000
iOS App Development (Swift)	90,000
Backend & API Development	70,000
Database Setup	15,000
Testing & QA (both platforms)	25,000
Deployment (both stores)	10,000
Total One-Time Cost	315,000

6.3 Additional Cost — Monetization Features (either alternative)

Item	Amount (Tk.)
Payment Gateway Integration (bKash/Nagad/Card aggregator)	25,000
Business Analytics & CSR Dashboard	20,000
CSR Certificate / Report Generator (PDF export)	15,000
Subscription & Billing Management	20,000
Logistics / Pickup Coordination Module	25,000
Total Additional Cost	105,000

6.4 Total Initial Investment by Alternative

Component	Alternative 1 (Tk.)	Alternative 2 (Tk.)
Base App Development (Section 6.1 / 6.2)	203,000	315,000
Monetization Features (Section 6.3)	105,000	105,000
Total Initial Investment	308,000	420,000

Recurring operating cost (hosting, maintenance, support) rises from 104,000 Tk./yr to a base of 168,000 Tk./yr to support the business tier, before payment-gateway processing fees (~2.5% of monetized revenue) are added, for either alternative — both are carried into the Financial & Investment Analysis below.

7. Financial & Investment Analysis

Following the feasibility framework — Financial Analysis, then Investment Analysis, then a comparison of alternatives by evaluating both development alternatives from Section 6 rather than the recommended path alone. Each year's projected net cash flow (F) is discounted back to today's value using the Present Value formula:

$$P = F / (1 + I)^n$$

where P = present value, F = net cash flow in year n, I = discount rate (12%), and n = year number. For example, Year 1's net cash flow of 144,500 Tk. discounts to $P_1 = 144,500 / (1.12)^1 \approx 129,018$ Tk.

Both alternatives deliver the identical FoodShare BD feature set and business model to end users — Flutter vs. native is an implementation choice, not a difference in what donors, recipients, or business accounts see or pay for. Revenue and operating-cost projections are therefore treated as identical across both alternatives; the two paths differ only in upfront development investment (308,000 Tk. vs. 420,000 Tk., Section 6.4). The shared cash flow below is the basis for both alternatives' investment analysis.

7.1 Revenue & Cash Flow Projection (Shared Basis)

Revenue is built from three streams — Business Pro subscribers, Corporate Partner accounts, and paid pickup requests — net of operating cost and payment-gateway processing fees (~2.5% of monetized revenue):

A) Financial Analysis for Alternative 1 — Cross-Platform Development (Flutter), with monetization live

- **Initial Cost:** 308,000 Tk. (203,000 Tk. Flutter base app + 105,000 Tk. monetization features)
- **Benefits and Costs (Yearly Basis — Year 1)**
 - Business Pro Subscriptions (15 × 10,000 Tk.): 150,000 Tk.
 - Corporate CSR Partnerships (2 × 60,000 Tk.): 120,000 Tk.
 - Pickup-as-a-Service Fees (400 × 75 Tk.): 30,000 Tk.
 - Recurring Operating Cost (hosting, maintenance, support): 148,000 Tk.
 - Payment Gateway Fees (~2.5% of 300,000 Tk. revenue): 7,500 Tk.

Net Return Per Year = 150,000 + 120,000 + 30,000 – 148,000 – 7,500
= 144,500 Tk.

Year	Net Flow F (Tk.)	PV @12% (Tk.)	Cum. PV (Tk.)
1	144,500	129,018	129,018
2	144,500	115195	244213
3	144,500	102852	347065
4	144,500	91832	438897
5	144,500	81993	520890

In the Alternative 1:

- Net Investment : 308,000 Tk.
- System Life cycle : 5 years
- Pay Back Period : 3 years
- Net present return value : 347,065 Tk.

By Year 5, cumulative present value reaches 520,890 Tk. against a 308,000 Tk. investment — Alternative 1 clears its cost comfortably within the 5-year system life.

B) Financial Analysis for Alternative 2 — Native Development (Android + iOS separately), with monetization live

- **Initial Cost:** 420,000 Tk. (315,000 Tk. native build + 105,000 Tk. monetization features)
- **Benefits and Costs (Yearly Basis)**
 - Business Pro Subscriptions (15 × 10,000 Tk.): 150,000 Tk.
 - Corporate CSR Partnerships (2 × 60,000 Tk.): 120,000 Tk.
 - Pickup-as-a-Service Fees (400 × 75 Tk.): 30,000 Tk.
 - Native Performance & Store-Ranking Premium (better UX drives higher conversion/retention): 50,000 Tk.
 - Recurring Operating Cost: 168,000 Tk.
 - Payment Gateway Fees (~2.5% of 350,000 Tk. revenue): 8,750 Tk.

**Net Return Per Year = 150,000+120,000+30,000+50,000-168,000-8,750
= 173,250 Tk.**

Year	Net Flow F (Tk.)	PV @12% (Tk.)	Cum. PV (Tk.)
1	173,250	154868	154688
2	173,250	138114	292802
3	173,250	123316	416118
4	173,250	110104	526222
5	173,250	98307	624529

In the Alternative 2:

- Net Investment : 420,000 Tk.
- System Life cycle : 5 years
- Pay Back Period : 4 years
- Net present return value : 526,222 Tk.

Alternative 2 needs 112,000 Tk. more capital and one extra year to pay back — but converts that into a higher total return: 624,529 Tk. by Year 5 versus 520,890 Tk. for Alternative 1, a genuine trade-off rather than a straightforward loss.

7.2 Comparison of Alternatives

Feature	Alternative 1	Alternative 2
Development Approach	Cross-Platform (Flutter)	Native (Android + iOS)
Initial Investment	308,000 Tk.	420,000 Tk.
System Life	5 years	5 years
Payback Period	3 years	4 years
Net present return value	347,065 Tk.	526,222 Tk.
5-Year Net Present Value	520890 Tk.	624529 Tk.

Neither alternative dominates outright. Alternative 1 recovers its investment a full year sooner on 112,000 Tk. less capital; Alternative 2 returns 103,639 Tk. more over the 5-year system life, at the cost of that extra capital and the extra year of waiting. Section 9 recommends Alternative 1, in line with the payback-focused objective set in Section 3..

8. Feasibility Remarks

- **Technical:** Flutter + Firebase/Node.js remains sufficient; the new scope (payments, dashboard, subscriptions) is standard functionality with mature libraries and a payment aggregator, not custom infrastructure.
- **Economic:** Alternative 1 pays back in 3 years with a Net Present Return Value of 520,890 Tk. by Year 5 — financially sound on its own. Section 7.2 shows that speed comes at the cost of 103,639 Tk. less total return than Alternative 2, a genuine trade-off rather than a clear win on every measure.
- **Operational:** Revenue depends on converting business donors to paying tiers — recommend piloting the Business Pro tier with 5–10 restaurants in Dhaka to validate pricing and conversion before wider rollout.

9. Recommendation

Proceed with Alternative 1 (Cross-Platform, Flutter) plus the monetization features in Section 6.3, for a total initial investment of 308,000 Tk. Section 7.2 shows this isn't a clean dominance — Alternative 2 returns 103,639 Tk. more over 5 years — but it needs 112,000 Tk. more capital upfront and a full year longer to recover it. Given the project's objective (Section 3) of reaching positive cash flow within a reasonable payback period, Alternative 1's 3-year payback makes it the better fit for a first business-model pivot; Alternative 2 remains worth revisiting later if maximizing total return matters more than speed of capital recovery.

10. Conclusion

Adding a business model changes FoodShare BD from a grant-dependent donation tool into a self-funding platform, without charging the recipients it serves. The cost and investment findings above support moving forward with a pilot build, followed by a Business Pro pilot with a small group of Dhaka restaurants to validate the revenue assumptions before scaling further.